



INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

ASSIGNMENT

Assignment Title: Linux System Administration Adventure

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Course: Foundation of Linux, Networking and Bash Scripting

Course Code: BVWS101

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Exercise 1

```
(cyberfarmer@kali)-[~]  
$ sudo touch assignment.sh  
[sudo] password for cyberfarmer:  
  
(cyberfarmer@kali)-[~]  
$ sudo nano assignment.sh  
  
(cyberfarmer@kali)-[~]  
$ sudo bash assignment.sh  
Enter your name:Garba  
Hello, Garba Welcome!
```

```
cyberfarmer@kali: ~  
GNU nano 8.6 assignment.sh *  
#!/bin/bash  
read -p "Enter your name:" name  
echo "Hello, $name Welcome!"  
|
```

Exercise 2

```
read -p "Enter filename: " filename  
if [ -f "$filename" ]; then  
    echo "File '$filename' exists."  
else  
    echo "File '$filename' not found."  
fi
```

```
(cyberfarmer@kali)-[~]  
$ sudo nano assignment.sh  
  
(cyberfarmer@kali)-[~]  
$ sudo bash assignment.sh  
Enter filename: Records  
File 'Records' not found.
```

Exercise 3

```
GNU nano 8.6 assignment.sh  
#!/bin/bash  
echo "The current date and time is: $(date)"
```

```
(cyberfarmer@kali)-[~]  
$ sudo bash assignment.sh  
The current date and time is: Wed 29 Oct 2025 11:42:22 WAT
```

Exercise 4

```
#!/bin/bash
# Prompt user for two numbers
read -p "Enter first number: " a
read -p "Enter second number: " b
# Prompt user for operation
echo "Choose and operation"
echo "1) Addition (+)"
echo "2) Subtraction (-)"
echo "3) Multiplication (*)"
echo "4) Division (/)"
read -p "Enter your choice (1-4) " op
case $op in
    add) result=$((a + b));;
    subtract) result=$((a - b));;
    multiply) result=$((a * b));;
    divide) result=$((a / b));;
    *) echo "Invalid operation"; exit 1;;
esac
echo "Result: $result"
```

```
(cyberfarmer@kali)-[~]
└─$ sudo bash assignment.sh
[sudo] password for cyberfarmer:
Enter first number: 2
Enter second number: 4
Choose and operation
1) Addition (+)
2) Subtraction (-)
3) Multiplication (*)
4) Division (/)
Enter your choice (1-4) 2
Invalid operation
```

Exercise 5

```
#!/bin/bash
# Ask for a directory path
echo "Enter directory path:"
read dir

# Count only files (not folders)
count=$(find "$dir" -maxdepth 1 -type f | wc -l)
echo "Number of files: $count"
```

```
(cyberfarmer@kali)-[~]
└─$ pwd
/home/cyberfarmer

(cyberfarmer@kali)-[~]
└─$ sudo bash assignment.sh
Enter directory path:
/home/cyberfarmer
Number of files: 12
```

Exercise 6

```
#!/bin/bash
# Ask for a username
echo "Enter username:"
read user

# Check if the user is logged in
if who | grep -w "$user" > /dev/null; then
    echo "$user is logged in."
else
    echo "$user is not logged in."
fi
```

```
(cyberfarmer@kali)-[~]
$ sudo bash assignment.sh
Enter username:
cyberfarmer
cyberfarmer is logged in.
```

Exercise 7

```
#!/bin/bash
# Ask for two numbers
echo "Enter first number:"
read a
echo "Enter second number:"
read b

# Compare the numbers
if [ "$a" -gt "$b" ]; then
    echo "$a is greater"
elif [ "$a" -lt "$b" ]; then
    echo "$b is greater"
else
    echo "Both numbers are equal"
fi
```

```
(cyberfarmer@kali)-[~]
$ sudo bash assignment.sh
Enter first number:
7
Enter second number:
4
7 is greater
```

Exercise 8

```
#!/bin/bash

# Start with count = 1
count=1

# Loop while count is less than or equal to 10
while [ $count -le 10 ]; do
    echo "Loop $count"
    count=$((count ++))
done
```

```
(cyberfarmer@kali)-[~]
└─$ sudo bash assignment.sh
Loop 1
Loop 2
Loop 3
Loop 4
Loop 5
Loop 6
Loop 7
Loop 8
Loop 9
Loop 10
```

Exercise 9

```
#!/bin/bash
# Show system info
echo "Hostname: $(hostname)"
echo "Kernel Version: $(uname -r)"
echo "Uptime: $(uptime -p)"
```

```
(cyberfarmer@kali)-[~]
└─$ sudo bash assignment.sh
Hostname: kali
Kernel Version: 6.16.8+kali-amd64
Uptime: up 1 hour, 31 minutes
```

Exercise 10

```
#!/bin/bash
# Ask for a filename
echo "Enter filename to backup:"
read file

# Copy the file with .bak extension
cp "$file" "$file.bak"
echo "Backup created: $file.bak"
```

```
(cyberfarmer@kali)-[~]
└─$ sudo bash assignment.sh
Enter filename to backup:
ahmed
Backup created: ahmed.bak
```

Exercise 11

```
#!/bin/bash
# Ping Google to check internet
ping -c 1 google.com > /dev/null

# Show result
if [ $? -eq 0 ]; then
    echo "Internet is working."
else
    echo "No internet connection."
fi
```

```
(cyberfarmer@kali)-[~]
$ sudo bash assignment.sh
Internet is working.
```

Exercise 12

```
#!/bin/bash
# Show disk usage of current folder
echo "Disk usage of current directory:"
du -sh
```

```
(cyberfarmer@kali)-[~]
$ sudo bash assignment.sh
Disk usage of current directory:
21M .
```

Exercise 13

```
#!/bin/bash
# Ask the user for a number
echo "Enter a number:"
read n

# Start from 2 and go up to n, step by 2
i=2
while [ $i -le $n ]
do
    echo $i
    i=$((i + 2))
done
```

```
(cyberfarmer@kali)-[~]
$ sudo bash assignment.sh
Enter a number:
10
2
4
6
8
10
```

Exercise 14

```
#!/bin/bash
# Ask for a filename
echo "Enter filename:"
read file

# Check permissions
[ -r "$file" ] && echo "File is readable"
[ -w "$file" ] && echo "File is writable"
[ -x "$file" ] && echo "File is executable"
```

```
(cyberfarmer@kali)-[~]
$ sudo bash assignment.sh
Enter filename:
ahmed
File is readable
File is writable
```

Exercise 15

```
#!/bin/bash
# Show memory usage
free -h
```

```
(cyberfarmer@kali)-[~]
$ sudo bash assignment.sh
```

	total	used	free	shared	buff/cache	available
Mem:	1.9Gi	1.3Gi	177Mi	11Mi	643Mi	637Mi
Swap:	2.0Gi	8.0Ki	2.0Gi			

Exercise 16

```
#!/bin/bash
# Show who is logged in
who
```

```
(cyberfarmer@kali)-[~]
$ sudo bash assignment.sh
cyberfarmer seat0      2025-10-29 10:08
cyberfarmer tty2       2025-10-29 10:08
```

Exercise 17

```
#!/bin/bash
# Ask for countdown time
echo "Enter countdown time in seconds:"
read secs

# Countdown loop
while [ $secs -gt 0 ]; do
    echo "$secs"
    sleep 1
    ((secs--))
done
echo "Time's up!"
```

```
(cyberfarmer@kali)-[~]  
$ sudo bash assignment.sh  
Enter countdown time in seconds:  
10  
10  
9  
8  
7  
6  
5  
4  
3  
2  
1  
Time's up!
```

Exercise 18

```
#!/bin/bash  
# Ask for a filename  
echo "Enter filename:"  
read file  
  
# Count lines in the file  
lines=$(wc -l < "$file")  
echo "Number of lines: $lines"
```

```
(cyberfarmer@kali)-[~]  
$ sudo bash assignment.sh  
Enter filename:  
ahmed  
Number of lines: 0
```

Exercise 19

```
#!/bin/bash  
# Set the correct password  
password="letmein"  
  
# Ask for password  
echo "Enter password:"  
read input  
  
# Check if it matches  
if [ "$input" = "$password" ]; then  
    echo "Access granted"  
else  
    echo "Access denied"  
fi
```

```
(cyberfarmer@kali)-[~]  
$ sudo bash assignment.sh  
Enter password:  
letmein  
Access granted
```

Exercise 20

```
#!/bin/bash
# Ask for main folder name
echo "Enter main directory name:"
read main

# Create folder with subfolders
mkdir -p "$main"/{docs,images,scripts}
echo "Directory structure created successfully."
```

```
(cyberfarmer@kali)-[~]
└─$ sudo bash assignment.sh
Enter main directory name:
Documents
Directory structure created successfully.
```